

VENKATA ABHINAV PADALA

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EDUCATION

University of Massachusetts Lowell Master of Science in Computer Science; <i>Coursework: Algorithms, Artificial Intelligence, Operating Systems, Advanced Database Management</i>	Lowell, MA Aug 2024 – May 2026
SRM University Bachelor of Technology in Computer Science; <i>Coursework: AI & ML, DBMS, Python, Web Development, NLP, Deep Learning</i>	Vijayawada, India Sep 2020 – May 2024

TECHNICAL SKILLS

- **Languages:** Python, Java, JavaScript, C, C++, HTML, CSS, SQL
- **Data Science:** Regression Analysis, Data Mining, Dimension Reduction (PCA & Clustering), Random Forest, NLP, Neural Networks
- **Database/RDBMS:** MSSQL, PostgreSQL, MySQL, Oracle DB, Snowflake, MongoDB
- **Frameworks & Technologies:** .Net 6.0/5.0, ASP.Net MVC, Azure, AWS (Lambda, S3, DynamoDB, EC2, IAM), REST API, Git, Docker, Apache, SSIS, SSRS, Agile
- **Tools:** Automation, Matlab, Jupyter Notebook, Pycharm

EXPERIENCE

Software Engineering Intern , <i>nirvanaHealth RxAdvance, Southborough, MA</i>	Feb 2025 – Present
• Designed and implemented scalable healthcare microservices supporting 10K+ daily prescription transactions, building backend features and REST APIs within a production PBM platform. • Developed automated integration and regression testing pipelines that reduced production defects by 30% and improved system reliability across distributed services. • Architected and optimized CI/CD workflows, decreasing deployment time by 40% and accelerating release cycles through streamlined build and deployment automation. • Improved system efficiency and security by optimizing cloud resource utilization and enforcing RBAC and secure configuration management in a HIPAA-compliant environment.	
Prototyping & Lab Operations Engineer , <i>UMass Lowell Innovation Hub (M2D2), Lowell, MA</i>	Sep 2024 – Jan 2025
• Led end-to-end prototyping and 3D printing workflows for med-tech startups, translating design requirements into manufacturable specifications using SolidWorks and GrabCAD Print. • Provided technical troubleshooting and lab operations support, improving equipment turnaround time and reducing print failures through optimized geometry validation and material calibration. • Collaborated with cross-functional teams to bridge hardware and software constraints, ensuring production-ready prototypes aligned with startup and research requirements.	
Graduate Assistant (Summer) , <i>Hospitality & Event Services, UMass Lowell</i>	May 2025 – Sep 2025
• Orchestrated complex operational workflows for university housing services, acting as the primary point of contact for resolving logistical bottlenecks and ensuring business continuity. • Directed and mentored administrative staff, leading from the front to standardize data protocols and improved team workflow efficiency.	

PROJECTS

AI Self-Healing Automation JavaScript / Node.js, Llama 3.2, Ollama, Playwright	Jan 2026
• Built a resilient testing engine using Playwright and Ollama (Llama 3.2) that autonomously repairs broken CSS selectors in real-time. • Engineered a privacy-centric inference layer utilizing custom Regex sanitization and sequential execution to mitigate LLM hallucinations while ensuring 100% local data privacy and zero API costs. • Ensured cross-browser stability (Chromium, Firefox, WebKit) through automated execution dashboards. • Designed modular architecture with fail-safe validation logic to prevent incorrect auto-repairs, improving system reliability and reducing manual QA intervention.	
NYC Data Pipeline Automation and Optimization Python, Snowflake, PySpark	Mar 2025
• Architected a distributed ETL pipeline using PySpark to process multi-million row transit datasets with parallelized transformations and OpenWeatherMap API real-time metrics. • Engineered scalable Big Data processing workflows using PySpark to clean, join, and aggregate massive datasets, utilizing Python for ingestion and Snowflake for storage.	
Railway Reservation System Full-Stack Application (ASP.NET Core, Java)	Feb 2024
• Designed a layered full-stack architecture using ASP.NET Core and Java, separating presentation, business logic, and data access layers for maintainability and scalability. • Optimized database query performance using indexed search patterns and structured schema design, reducing response latency for booking and availability checks. • Implemented secure authentication, role-based access control, and input validation to get real-world production-grade system behavior.	

ACHIEVEMENTS & PUBLICATIONS

- Padala, Abhinav., Nagothu, S, Lokesh **Federated Learning with Differential Privacy for Edge AI.** (2025). Authored a research paper analyzing the convergence of Federated Learning and Differential Privacy to mitigate gradient leakage attacks and also optimizing communication efficiency in resource-constrained Edge AI ecosystems. [\[View Paper\]](#)
- **Selected Fellow, UMass LIFT Program.** (Fall 2025). Finalist in a seasonal competitive medtech entrepreneur fellowship. Developed "CareBridge," an AI-powered cognitive care platform, and devised commercialization strategies for aging solutions. [\[View Certificate\]](#)